Brain Tumor Detection in MRI Using Classical Image Processing: A Feature-Based Thresholding Approach

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Abstract- The main modality for brain tumor diagnosis is Magnetic Resonance Imaging (MRI) but manual reading is time-consuming and requires the availability of experts. In this paper, a series of classical Image and Video Processing (IVP) techniques are examined to determine how much the techniques can achieve without employing any learned model to separate tumorbearing from tumor-free slices of the brain MRI. We create a pipeline of grayscale conversion, resizing to 256×256, CLAHE (Contrast Limited Adaptive Histogram Equalization), Gaussian filtering and Canny edge detection and then Otsu thresholding, morphological opening and closing, and connected-component analysis. Each image has 5 global features extracted by hand and decision thresholds are optimized for a calibration split only. Experiments are conducted on the public Kaggle Brain Tumor MRI dataset with 7,200 images from which 4,480 images are used for calibration, 1,120 are used for validation and 1,600 are used as held out test images (glioma, meningioma, pituitary and notumor). The first rule, based on the segmented foreground fraction, achieves 74.94% test accuracy and makes all normal scans a tumour class prediction (0% specificity — same as the majority class baseline). Texture and contrast features discriminate much better, and a combined rule based on the z-score average of intensity standard deviation and Laplacian variance gives 82.06% accuracy, 84.82% precision, 92.67% recall, 88.57% F1-score and 50.25% specificity for the test set. The results indicate that classical IVP offers interpretable and low cost tumor cues, however it has limited specificity on normal scans and is therefore more appropriate for supporting screening and visualisation, rather than diagnosis. Keywords- brain tumor detection, MRI, image processing, CLAHE, Otsu thresholding, Canny edge detection, mathematical morphology, connected components, features Laplacian variance, handcrafted

I. Introduction

Brain tumor is an abnormal growth of cells in the brain or tissues around the brain. They can grow slowly and harmless, or fast-growing and dangerous, and the WHO makes classifications based on how they appear [1]. Gliomas develop from the glia cells that surround and support the brain and spinal cord; meningiomas grow on the membranes (meninges) that cover the brain and spinal cord; pituitary tumors form from the pituitary gland, located at the base of the brain. Early detection has a significant impact on treatment planning and outcomes for patients.

MRI is the imaging modality of choice in the brain because it provides a high contrast between soft tissues without using any radiation. However, there are many slices in a single study and only a trained radiologist can read the slices. If specialists are not available, interpretation is postponed. Computer-aided detection can lessen this burden by alerting about suspicious scans for special attention.

The majority of the recent research uses deep convolutional neural networks (CNNs) for this purpose [5]–[8]. They are accurate, but require significant amounts of labeled data, GPUs, and fine tuning, and the decisions are difficult to explain. Classical IVP methods, such as contrast enhancement, filtering, edge detection, thresholding and morphology [9] can be applied with minimal time, without the need for training and with the possibility of checking every intermediate step. They don't typically test their limits on real MRI data, on a held-out test set.

* In this paper, a complete classical IVP pipeline for brain MRI is developed and simple handcrafted features are used to classify tumour / no tumour images with a strict calibration/validation/test protocol, and the validation results are reported. The main contributions are:

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* Results via resizing, CLAHE, Gaussian filtering, Canny edges, and Otsu processed images are reproducible in an eight-stage pipeline.

* 7200 MRI slices were subjected to thresholding, morphology, connected components.

* A side by side comparison of basic IVP operations using MRI: point transforms, spatial filters, frequency domain filtering, edge detectors and thresholding methods.

* Five handcrafted features are compared quantitatively, thresholds are set using calibration data and tested using unseen data.

* An error analysis detailing why the rules for segmentation are not successful on normal scans, and why texture features (intensity spread and Laplacian variance) are more successful.

The rest of the paper is organized as follows. In Section II, related work is reviewed. The dataset description is covered in Section III. The methodology is described in detail in Section IV and the experimental setup in Section V. Section VI reports and discusses results, Section VII concludes.

II. LITERATURE REVIEW

Gordillo et al. [2] review the classical methods used for analyzing brain tumors in MRI. Their categorize method into threshold based, region based, edgebased and classification based. Simple and fast threshold methods like Otsu's criterion [10] have problems when the tumor intensities are similar to the healthy tissue intensities. Edge-based methods, such as the Canny detector [11] are suitable for detecting boundaries with a high degree of accuracy but are sensitive to all anatomical boundaries, including tumor margins. To clean binary masks, one is usually resorted to mathematical morphology [14] to remove small specks and fill gaps.

These days, learning-based methods are prevailing. Cheng et al. [4] showed how to classify glioma, meningioma and pituitary tumors by enhancing the tumor region and manually designing features to be used by a classifier. In the work of Sajjad et al. [5] the authors used a deep CNN with heavily augmented data to grade gliomas. Transfer learning from Image Net pretrained networks was used by Deepak and Ameer [6] and by Swati et al. [7] for the same three tumor types, and Badža and Barjaktarović [8] created a compact CNN for the same three tumor types. These techniques are highly accurate but need training data and GPU resources, and have limited interpretability.

This paper fills an important gap, providing a clear, quantitative baseline for simply classical IVP detection for a large modern dataset. The majority of classical studies report results on a few sample images or tune thresholds to the same data they evaluate. In this case each threshold is picked based on a calibration split of data, tested on a validation split and reported on a held-out test set, thus reporting the true threshold values which the features capture.

III. DATASET AND SYSTEM DESIGN

A. Dataset Description

Kagglehub library is used to download the publicly available Brain Tumor MRI Dataset by Nickparvar [16]. The version available includes 7,200 2 dimensional brain MR images in JPEG format, in 2 folders: one Training folder containing 4 class sub-folders and one Testing folder containing 4 class sub-folders. The balance between classes for both folders is just right (Table I). A sample glioma image (Tr-gl_1260.jpg) is 512 x 512 pixels, unsigned integer, 8-bit, intensities range from 0 to 252.

TABLE I. IMAGE COUNTS PER CLASS

Class	Training	Testing	Total	Label
Glioma	1,400	400	1,800	1 (Tumor)
Meningioma	1,400	400	1,800	1 (Tumor)
Pituitary	1,400	400	1,800	1 (Tumor)
No tumor	1,400	400	1,800	0 (No Tumor)
Total	**5,600**	**1,600**	**7,200**	

B. Binary Labeling

The study is focused on binary detection, and not tumour-type classification. The images in the notumor folder are labeled as 0, while the images in the three tumor folders are labeled as 1. Thus, 75% of the images are tumorlabeled. This imbalance is significant with respect to evaluation: a trival rule that always predicts "tumor" already achieves a 75% accuracy. For this reason we report specificity and balanced accuracy in addition to accuracy.

C. Data Partitioning

Shuffling is performed on the original Training folder (5,600 images) with the fixed random seed of 42, and then splitting is done using an 80/20 ratio of the calibration set and the validation set. Shuffling is not done on the original Testing folder (1,600 images). Feature and rule selection is done using the thresholds determined from the calibration set while feature comparison is done using the validation set.

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TABLE II. DATA PARTITIONS USED IN THE EXPERIMENTS

Partition	Source	Images	No tumor	Tumor
Calibration	80% of Training	4,480	1,133	3,347
Validation	20% of Training	1,120	267	853
Test	Testing folder	1,600	400	1,200

In the calibration data set, the class distribution after shuffling is 1,125 gliomas, 1,093 meningiomas, 1,129 pituitary tumors, and 1,133 no-tumors.

IV. PROPOSED METHODOLOGY

There are four phases involved: (A) exploratory analysis of basic IVPs on MRIs, (B) a consistent processing chain used on all images, (C) selection of hand-crafted features, and (D) threshold-based decision-making on calibrated data..

A. Exploratory Study of IVP Operations

The following step entailed running a standard IVP operation for each of the methods on a randomly chosen glioma slice scaled to 256 × 256 pixels before pipeline repair.

1) Histogram processing. The histogram of the slice in grayscale representation shows an overwhelming peak around zero because of the dark background. Histogram equalization has the effect of distributing the intensity values across the entire range of 0 to 255, but it has the disadvantage of adding noise from the background.

2) Point (intensity) transformations. There were five transformations in which r represented the input intensity while s represented the output.:

* Digital negative: s = 255 − r.
* Contrast stretching between the 2nd and 98th percentiles p2 and p98: s = 255 (r − p2) / (p98 − p2), clipped to [0, 255].
* Intensity-level slicing: pixels with 100 ≤ r ≤ 200 are set to 255, all others to 0.
* Logarithmic transformation, which expands dark intensities, given in (1).
* Power-law (gamma) transformation with γ = 0.7, which brightens mid-tones, given in (2).

$$s = 255 \cdot \log(1 + r) / \log(256) \quad (1)$$

$$s = 255 \cdot (r / 255)^{\gamma} \quad (2)$$

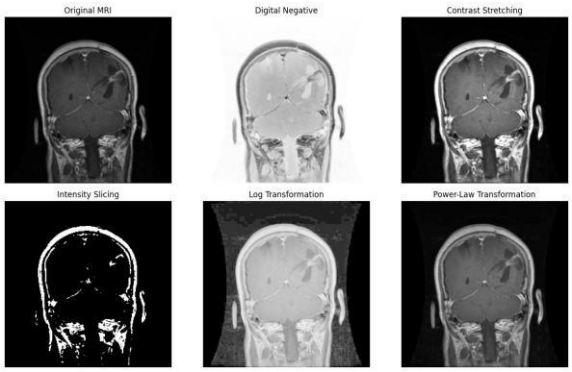


Fig. 1. Point transformations applied to a 256 × 256 glioma slice: original, digital negative, contrast stretching, intensity slicing, log and power-law (γ = 0.7).

3) Spatial filtering. The three filters that were used in order to smooth were 5 x 5 average filter, 5 x 5 Gaussian filter (σ = 1.0), and 5 x 5 median filter. For sharpening, the 3 x 3 kernel [[0, -1, 0], [-1, 5, -1], [0, -1, 0]] was used. This was because it adds negative Laplacian to the image.

4) Frequency-domain filtering. The DFT of the image was done in 2D and the shift is performed to make sure that the zero frequency is in the center.

Spectrum magnitude was plotted using 20 log(|F(u, v)| + 1). Ideal low pass and high pass filters with cutoff radius of D0 = 25 pixels were used, as in Equations (3) and (4).

$$H_LP(u, v) = 1 \text{ if } D(u, v) \le D0, \text{ otherwise } 0 \quad (3)$$

$$H_HP(u, v) = 1 - H_LP(u, v) \quad (4)$$

The inverse DFT of the low-pass result is the blurred outline of the brain; the high-pass result has edges and texture only. Both results have concentric rings of artifacts, due to the sharp cut-off of the ideal filter.

5) Edge detection. Following Gaussian smoothing, the Sobel operator (3 x 3) yielded G = √(Gx2 + Gy2). The Canny edge detector [11] using hysteresis thresholds of 50 and 150 generated thin and connected edges and thus was chosen for the pipeline.

6) Thresholding and segmentation. Otsu's technique [10] determines the optimal threshold t* which maximizes the between-class variance of the histogram given by equation (5), where ω0, ω1 are the class probabilities and μ0, μ1 are the class means.

$$t^* = \arg\max_t \; \omega0(t)\, \omega1(t)\, [\mu0(t) - \mu1(t)]^2 \quad (5)$$

For the sample slice, Otsu chose t* = 39. Additionally, Adaptive Gaussian thresholding method (block size = 21, constant C = 3) was tried; it is influenced by local texture at all points and resulted in fragmented mask, therefore Otsu thresholding globally was used.

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7) Morphology and connected components. The elliptical structuring element of size 3 × 3 was employed for erosion, dilation, opening (erosion followed by dilation, eliminating small spots), and closing (dilation followed by erosion, filling small holes). The connected component labeling process using 8 connectivity found the areas, bounding boxes, and centroids of each of the foreground objects. In the test slice, there were 5 components in the closed Otsu mask, one of which covered the brain having an area of 27,113 pixels (155 × 220 bounding box).

B. Final Processing Pipeline

Each image goes through eight stages:

* Loading as grayscale image using OpenCV library with (cv2.IMREAD_GRAYSCALE) flag.

  * Resize to 256 x 256 pixels using area interpolation (INTER_AREA), which avoid aliasing during downsampling.

* CLAHE with clip limit 2.0 using 8 x 8 tile grid to enhance contrast locally without increasing noise [12].

* Gaussian smoothing with 5x5 kernel and σ = 1.0 for noise removal before edge detection and thresholding.

* Edge detection using Canny with thresholds 50 and 150.

* Otsu thresholding of the smoothed image to obtain binary mask of foreground.

* Morphological cleaning using first opening, then closing with 3x3 elliptical structuring element.

* Connected components analysis using 8-connectivity, and saving area, bounding box and centroid of each connected component. The largest component is selected as candidate region and can be marked in red color on the original slice image.

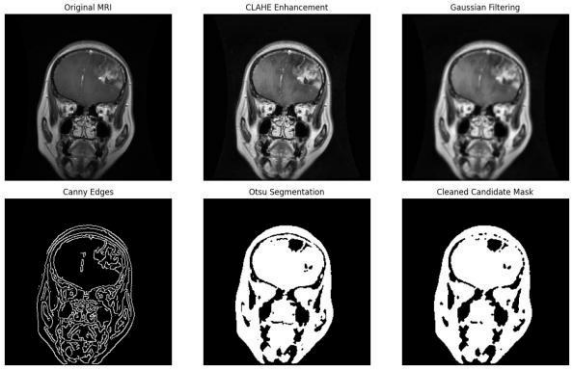


Fig. 2. Output of each pipeline stage for a calibration glioma image: original, CLAHE, Gaussian filtering, Canny edges, Otsu segmentation and the cleaned candidate mask (Otsu threshold = 67, one connected region).

C. Feature Extraction

There are five scalar quantities that are calculated from the preprocessed image (Table III). Consider the enhanced image E with N = 256 × 256 pixels, Otsu Mask M and Canny edges C.

TABLE III. HANDCRAFTED FEATURES

Feature	Definition	What it captures
Foreground fraction	Share of pixels with M > 0	Size of bright segmented region
Edge fraction	Share of pixels with C > 0	Boundary and fine structure
Mean intensity	mean(E) / 255	Overall brightness
Intensity std	std(E) / 255	Spread of intensities (contrast)
Laplacian variance	var(∇^2E), 64-bit float	Sharpness and fine texture

Laplacian variance [15], which is commonly used as a feature, is defined by equation (6).

$$V_L = (1/N)\ \Sigma_(x,y)\ [\nabla^2 E(x, y) - mean(\nabla^2 E)]^2 \quad (6)$$

For the baseline experiment, the foreground ratio is calculated from the cleaned image mask, while for the five feature analysis, it is calculated from the uncleaned image mask using the Otsu threshold. Also noted was the Otsu threshold value and the number of connected components.

D. Decision Rules

1) Baseline foreground rule. If the percentage of the foreground area of an image is less than a threshold t, then the image is considered to be that of a tumor. This procedure was repeated for one hundred equidistant values of t between the min and max calibration foreground percentage.

2) Single-feature rules. Threshold candidates for every feature f consist of the midpoints between subsequent values of calibrated feature sorted uniquely. Two possible ways (tumor for f ≥ t, tumor for f < t) are tested, and the optimal one is selected on the basis of the calibration accuracy

3) Combined rule. Two best performing measures, namely Intensity Standard Deviation and Laplacian Variance, are normalized based on the mean μ and standard deviation σ obtained from the calibration, and then their weighted average is calculated as S according to equation (7). A threshold for S and the sign are selected through the accuracy search

$$S = ½\ [\ z_std + z_lap\ ],\ \ z = (f - \mu) / \sigma \quad (7)$$

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4) Specificity-constrained rule. In exchange for better recall, a modified version of the combined strategy retains only the threshold values that have calibration specificity of at least 80% and chooses the threshold value with the best recall out of those.

E. Evaluation Metrics

When the tumor is considered as the positive class, TP, TN, FP, and FN stand for true positive, true negative, false positive, and false negative, respectively. Acc stands for accuracy, P represents precision, R is recall, and BA stands for balanced accuracy. These measures are shown in (8)-(13).

$$Acc = (TP + TN) / (TP + TN + FP + FN) \quad (8)$$

$$Precision = TP / (TP + FP) \quad (9)$$

$$Recall = TP / (TP + FN) \quad (10)$$

$$Specificity = TN / (TN + FP) \quad (11)$$

$$F1 = 2 \cdot P \cdot R / (P + R) \quad (12)$$

$$BA = (Recall + Specificity) / 2 \quad (13)$$

V. EXPERIMENTAL SETUP AND IMPLEMENTATION

All the experiments were performed using a Google Colab notebook with Python. Data was directly imported from the Kaggle cache using kagglehub. The use of a GPU or any model training is not necessary; everything is deterministic except the shuffled data that is seeded.

TABLE IV. SOFTWARE ENVIRONMENT

Component	Library / version	Role
Image processing	OpenCV 5.0.0 [17]	Resizing, CLAHE, filtering, DFT, Canny, Otsu, morphology, components
Numerical computing	NumPy 2.1.3	Arrays, point transforms, feature statistics
Evaluation	scikit-learn [18]	Accuracy, precision, recall, F1, confusion matrices
Data handling	pandas	Feature tables, summaries, CSV export
Visualization	Matplotlib	Stage figures, histograms, box plots

The steps involved in the experiment were as follows:

* Count the number of images per class and ensure that the folder structure is correct (Table I).

* Conduct the exploratory IVP operations on one sample slice (Section IV-A).

* Conduct the full pipeline on 20 images per class as the pilot sample to validate the behavior of features (80 images).

* Extract features from all 4480 calibration, 1120 validation, and 1600 test images.

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* Evaluate per-category accuracy, false positives, and false negatives; also visualize images that have been wrongly classified.

* Store results in CSV format for test image classification, metrics, and per-category analysis.

An interactive tool of smaller size was also implemented in the notebook. The user can upload an MRI slice in JPEG or PNG format. The pipeline works and displays the initial image, CLAHE, Gaussian blur, Canny edge detection, Otsu thresholding and mask cleaning steps, draws bounding boxes for five biggest areas with ≥100 pixels, and outputs the experimental prediction without diagnosing it medically.

VI. EXPERIMENTAL RESULTS AND EVALUATION

While the optimal rule using intensity SD and Laplacian variance achieves 82.06% classification accuracy and 50.25% specificity, the segmentation area rule is no better than guessing tumor every time. The reasoning behind both these conclusions is explained here.

A. Exploratory Feature Analysis

Preliminary experiments on 20 images per class (see Table V) revealed that the no-tumor class has the largest Otsu threshold, the largest edge ratio and the largest number of connected components, while its foreground ratio is nearly equal to that of the glioma and meningioma classes.

TABLE V. MEAN PIPELINE MEASUREMENTS ON THE PILOT SAMPLE

Class	Otsu thr.	Fg. fraction	Edge fraction	Comp.
Glioma	64.40	0.3854	0.0839	3.3
Meningioma	72.55	0.3963	0.0884	2.0
Pituitary	76.15	0.5014	0.1207	5.1
No tumor	83.45	0.3958	0.1323	7.3

In the total calibration dataset, the mean of the foreground fraction from the cleaned mask was 0.4276 for images without a tumor and 0.4420 for images with a tumor, a difference of only 0.0144. The edge

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fraction (0.1322 versus 0.0975) and component number (7.41 versus 3.95) showed more variation. The largest connected component in terms of the number of pixels for ten images each was 24,067 (36.7% of the image), 26,315 (40.2%), 26,937 (41.1%) and 32,562 (49.7%).

Explanation of Fig. 3 is provided to show why there is poor separation of the classes by area-based features. The result of the threshold according to Otsu splits the image into the tissue that appears bright and the background which is darker. As such, the biggest region of interest becomes the entire head and not the lesion, which in this case includes 20,530 pixels (31.33%).

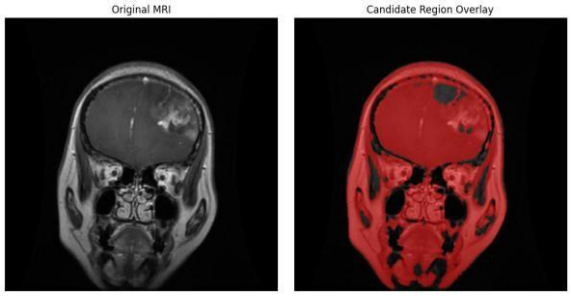


Fig. 3. Largest connected region (red) after Otsu thresholding and morphology on a tumor-labeled slice. The region covers the head and brain tissue rather than the tumor alone.

B. Baseline: Foreground-Fraction Rule

Thresholding gave us t = 0.6935 (tumor if foreground fraction < t), where the accuracy rate is 74.73%. Virtually every image had a foreground fraction less than 0.69, therefore, according to the classification rules, "tumor" would be the output for virtually everything. Accuracy rate on validation is 76.16% (852 out of 853 tumors were identified and only one of 267 normal cases were correctly classified).

TABLE VI. TEST CONFUSION MATRIX, FOREGROUND-FRACTION RULE

	Predicted no tumor	**Predicted tumor**
Actual no tumor (400)	0	400
Actual tumor (1,200)	1	1,199

This yields an accuracy of 74.94%, 74.98% precision, 99.92% recall and 85.67% F1-Score. While these numbers seem to be fair, they are actually quite misleading since specificity is 0%, false-positive rate is 100%, and the accuracy is just a little above that of simply predicting "tumor" all the time, i.e., 75%.

C. Comparison of Handcrafted Features

Comparison between class means in Table VII gives the difference in value of the respective feature in the calibration set for the two classes. For all features apart from foreground fraction, the value in tumor images is lesser than that in the normal cases. Of all features, the ratio is maximum for Laplacian variance (2.8).

TABLE VII. MEAN FEATURE VALUES ON THE CALIBRATION SET

Feature	**No tumor**	**Tumor**	**Abs. diff.**
Laplacian variance	2,750.95	980.81	1,770.13
Mean intensity	0.2999	0.2515	0.0484
Intensity std	0.2677	0.2293	0.0384
Edge fraction	0.1326	0.0975	0.0351
Fg. fraction (raw Otsu)	0.4244	0.4383	0.0140

Each feature was then represented by a threshold rule on the calibration set and tested on the validation set (Table VIII). Standard deviation of intensity and Laplacian variance are the most effective features, and both significantly increase the specificity compared to the baseline.

TABLE VIII. SINGLE-FEATURE RULES ON THE VALIDATION SET (%)

Feature	**Thr.**	**Acc.**	**Prec.**	**Rec.**	**Spec.**
Intensity std	0.2635	83.48	85.16	94.84	47.19
Laplacian var.	1,880.31	83.12	82.55	98.71	33.33
Edge fraction	0.1408	79.20	81.50	94.02	31.84
Mean intensity	0.3671	78.39	79.57	96.37	20.97
Fg. fraction	0.6902	76.16	76.21	99.88	0.37

In the test set, intensity-std criterion achieved an accuracy of 81.06%, precision of 85.45%, recall of 90.08% and specificity of 54.00%, with 184 false positives and 119 false negatives.

D. Combined Two-Feature Rule

The average of intensity std and Laplacian variance z-scores (IV-D), on the other hand, improved the accuracy up to 85.89% by yielding 97.30% tumor recall and 49.44% notumor specificity (Table IX).

TABLE IX. VALIDATION COMPARISON OF THE STRONGEST RULES (%)

Rule	**Accuracy**	**Tumor recall**	**Specificity**
Combined average	85.89	97.30	49.44
Intensity std	83.48	94.84	47.19
Laplacian variance	83.12	98.71	33.33

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In the out-of-sample test data, the combined rule had an accuracy of 82.06% (Table X), where it identified 201 out of 400 normal scans and 1,112 out of 1,200 tumor scans.

TABLE X. TEST CONFUSION MATRIX, COMBINED RULE

	Predicted no tumor	Predicted tumor
Actual no tumor (400)	201	199
Actual tumor (1,200)	88	1,112

E. Overall Test Performance

In Table XI, the rules are lined up next to each other, along with the easy majority class rule, based on 1,600 images used for testing. The balanced accuracy is the best single number in this case since it is not affected by the 3:1 class imbalance.

TABLE XI. TEST-SET PERFORMANCE OF ALL RULES (%)

Rule	Acc.	Prec.	Rec.	F1	Spec.	Bal.
Always "tumor"	75.00	75.00	100.00	85.71	0.00	50.00
Foreground fraction	74.94	74.98	99.92	85.67	0.00	49.96
Rule	**Acc.**	**Prec.**	**Rec.**	**F1**	**Spec.**	**Bal.**
Intensity std	81.06	85.45	90.08	87.71	54.00	72.04
Combined (std + Lap.)	**82.06**	**84.82**	**92.67**	**88.57**	**50.25**	**71.46**

While the combined rule provides the highest accuracy and F1-score, the intensity-std rule is more specific and provides balanced accuracy. The class-wise report also shows that the combined rule outperforms the baseline significantly in terms of metrics: for the combined rule, the no-tumor class provides 0.70 precision and 0.50 recall as opposed to 0.85 and 0.93 of the tumor class.

The specificity-constrained form of the combined rule with a calibration specificity of at least 80% had a specificity of 77.53%, but the tumor recall decreased to 64.83%. This reaffirms the difficulty in trading off between sensitivity and specificity for this set of features.

F. Per-Class Analysis

Table XII shows the breakdown of test accuracy based on the original class. The baseline model labels each picture as a tumor, hence its accuracy is 100% for all tumor classes but 0% for all normal pictures. The rule combination performs almost perfectly in identifying pituitary tumors, reasonably for gliomas, and relatively poorly for meningiomas.

TABLE XII. TEST ACCURACY BY CLASS (400 IMAGES EACH)

Class	Fg. rule correct	Fg. rule (%)	Comb. correct	Comb. (%)
Glioma	400	100.00	368	92.00
Meningioma	399	99.75	347	86.75
Pituitary	400	100.00	397	99.25
No tumor	0	0.00	201	50.25

The Table XIII gives the mean value of features per class for the test data. The images without tumors show the largest intensity range and Laplacian variance. Meningioma is nearest to no-tumor class in terms of intensity standard deviation (0.2490 vs. 0.2670).

TABLE XIII. MEAN FEATURE VALUES PER CLASS ON THE TEST SET

Class	Int. std	Lap. var.	Edge frac.	Mean int.
Glioma	0.2275	908.98	0.0913	0.2341
Meningioma	0.2490	807.04	0.0968	0.2682
Pituitary	0.2312	1,240.02	0.1187	0.3003
No tumor	0.2670	1,619.10	0.1221	0.2911

G. Discussion and Limitations

Why segmentation area fails. Global Otsu segmentation will differentiate the head from the background but not tumor from the brain. The mask obtained by applying global Otsu is between 36 and 50 percent of the image in all classes, making the size of the mask meaningless from a diagnostic point of view.

Why texture features work better. These two features characterize the entire slice. The normal slices in this database yield higher average values for both (Table VII). This may be because the normal slices have more intricate high contrast structures such as sulci, skull borders, and ventricles, while the tumor slices have larger uniform regions, which make the values of these two features smaller.

Possible dataset bias. This Kaggle dataset is formed from images that belong to a number of other public datasets [16]. The images without tumors are not all taken from the same database as those with tumors. Thus, some of the separation caused by global textures could result from scanning technology and not pathology. It should be tested using a single-database dataset.

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Further limitations. (i) The chosen areas are not true tumor segmentations; there is no ground truth in terms of pixels, so localization cannot be evaluated. (ii) The values of all the parameters (CLAHE clip level, Gaussian sigma, thresholds for Canny, structuring element dimensions) are set manually, without optimization. (iii) Only 2-D slices of random orientation and unknown MRI sequence are considered. (iv) A specificity of 50 percent means that every second healthy image will be marked; this number is too high for unattended clinical practice.

VII. CONCLUSION AND FUTURE WORK

A fully classical algorithmic pipeline has been constructed and tested for detecting brain tumor in MRI slices. With the help of CLAHE, Gaussian filter, Canny edge detection, Otsu's thresholding, morphological and connected component operations using 7,200 images and with optimal thresholds being selected only from calibration data, the best performing classifier – the z-score of intensity variance and Laplacian variance – performed with an accuracy of 82.06%, recall of 92.67%, F1-score of 88.57%, and specificity of 50.25% on 1,600 test images.

It should be noted that the research has also uncovered an unexpected finding in terms of the negative outcome. The classifier based on segmented foreground area produces 74.94% accuracy and 99.92% recall, but it classifies all normal scans as a tumor and thus achieves performance equivalent to a constant guess "tumor." This would not have been revealed without specifying other metrics.

Thus, the classical IVP method is helpful in performing transparent preprocessing, visualization, and rough selection, yet it cannot be used for robust lesion detection alone. Future extensions may include:

* Preprocessing using skull stripping and segmentation using region growing, watershed or active contours so that the selected regions contain lesions only, not the entire head.

* Describing textures using methods like GLCM features, Local Binary Patterns and energy of wavelets along with training of a classifier such as SVM or random forest.

* Classical IVP pipeline as preprocessing for CNNs, tested against results from deep learning approaches [5]–[8].

* A validation on a single-source dataset containing pixel-level tumor mask labels, like BraTS [3] dataset.

* Using the binary lesion detection to perform classification of different tumor types.

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